Acid–Base Disorders

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Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. **Answer keys are on the slides** (red answers) for Cases 1, 2, 3 and 4. I checked each one against the lecture's own formulas; **Case 3's key is wrong** (see §2.11).

🔗 **Bridge from earlier lectures:**

- **Respiratory Failure / ARDS (Kotb): type 2 (hypercapnic) failure = respiratory acidosis**. Same rule: **careful O_2 in COPD**; ventilate when pH < 7.0–7.25, mental deterioration or fatigue.
- **COPD / Asthma / Sleep Apnoea (Kotb):**
- chronic CO_2 retainers → **chronic compensated respiratory acidosis** (Case 2)
- **obesity hypoventilation** (slide 10)
- a **"normal" $PaCO_2$ in a tiring asthmatic** is ominous *[added]*
- **Pulmonary Vascular Disease (Kotb): PE → hypoxaemia → respiratory alkalosis**.
- **Adrenal 1 & 2:**
- **Conn's (hyperaldosteronism)** and **Cushing's** → **chloride-resistant metabolic alkalosis** (+ hypokalaemia)
- **Addison's** → **normal-AG metabolic acidosis** (+ hyperkalaemia)
- **Cardiogenic Shock (Reihan):** shock → **lactic acidosis** (HAGMA); **bicarbonate** is a rescue therapy for refractory shock.
- **Substance Use (Khrad): ethanol**, **methanol**, **ethylene glycol**, **salicylate** poisoning; **propylene glycol** (lorazepam infusion vehicle *[added]*).
- **Tuberculosis (Kotb): isoniazid (INH)** overdose → seizures → **lactic acidosis** (the "I" in MUDPILES).
- **Pituitary 3: ADH** and osmolarity control (slide 7).
- **GERD:** *[added]* repeated vomiting or NG suction → **chloride-responsive metabolic alkalosis** (Case 3).

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Renal anatomy (nephron) and functions incl. **HCO_3^- handling**	Kidney = metabolic arm of pH control
Knowledge	⭐ Acidosis vs acidaemia; buffers; **normal ABG ranges**	pH 7.35–7.45, $PaCO_2$ 35–45, HCO_3^- 22–26
Knowledge	⭐ Mechanisms, causes, features of the 4 primary disorders	MUDPILES / HARDASS; Cl-responsive vs Cl-resistant
Knowledge	Treatment: cause, O_2, ventilation, **bicarbonate, fomepizole/haemodialysis**	Toxic alcohols; Cl-responsive → saline
Skill	⭐ Analyse an ABG: **simple vs compensated vs mixed**	Case questions

Domain	Objective	Likely tested as
Skill	⭐ **Anion gap**	AG = Na − (Cl + HCO_3)
Skill	**Tic-Tac-Toe** grid and **SMORE**	Identify the primary disorder
Skill	⭐ **Compensation formulas** → appropriate or mixed	Winter's formula etc.

> ⚠️ **Coverage gap:** the **"Tic-Tac-Toe" grid** is named in the objectives but **never taught** on the slides. It is drawn in the added diagram (§2.9).

Topics (slide 3): respiratory acidosis → respiratory alkalosis → metabolic acidosis → metabolic alkalosis → patient approach.

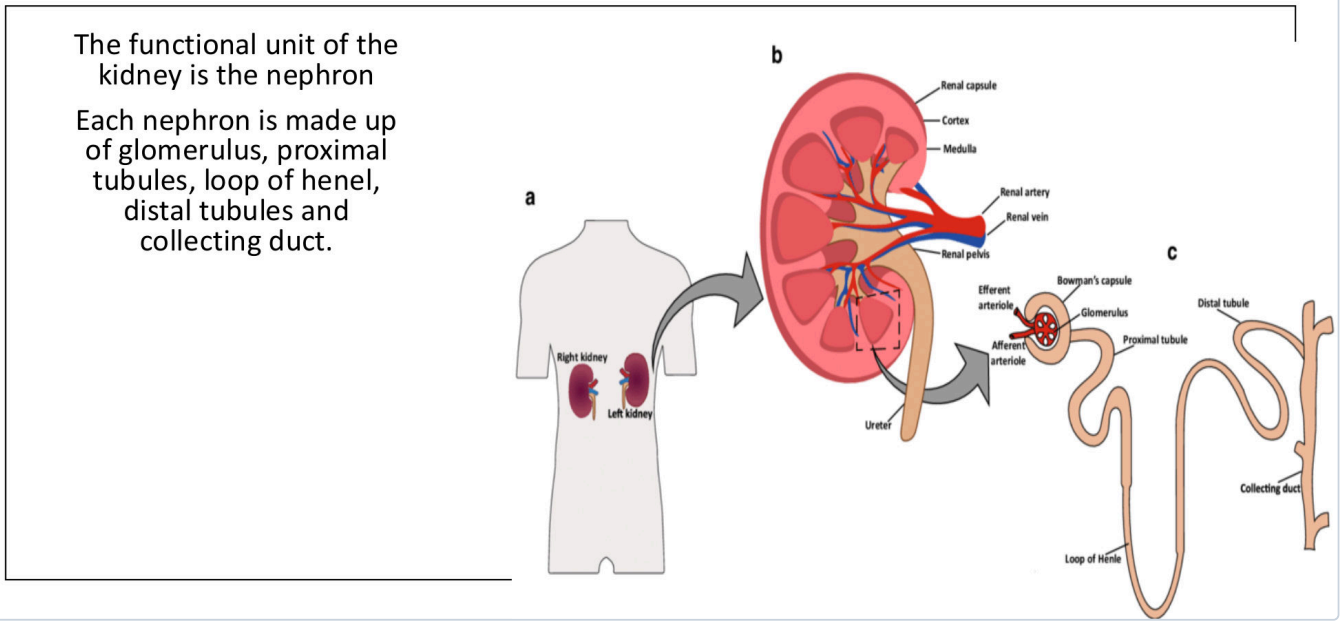
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2. Main Content (Lecture Order)

2.1 Renal Anatomy & Function (Slides 4–7)

Anatomy:

- **retroperitoneal**, either side of the spine at **T12–L3**
- the **right kidney sits slightly lower** (liver)
- outer **cortex**, inner **medulla**
- **Nephron** (functional unit): **glomerulus → proximal tubule → loop of Henle → distal tubule → collecting duct**



Slide 5: Kidney and nephron

Function	Examples
Excretory	Waste products
Regulatory	Fluid volume and composition (incl. **pH**)
Endocrine	**Erythropoietin, renin, angiotensin, prostaglandins**
Metabolic	**Vitamin D activation**; degradation of small proteins (e.g. **insulin**)

Volume and osmolarity (slide 7):

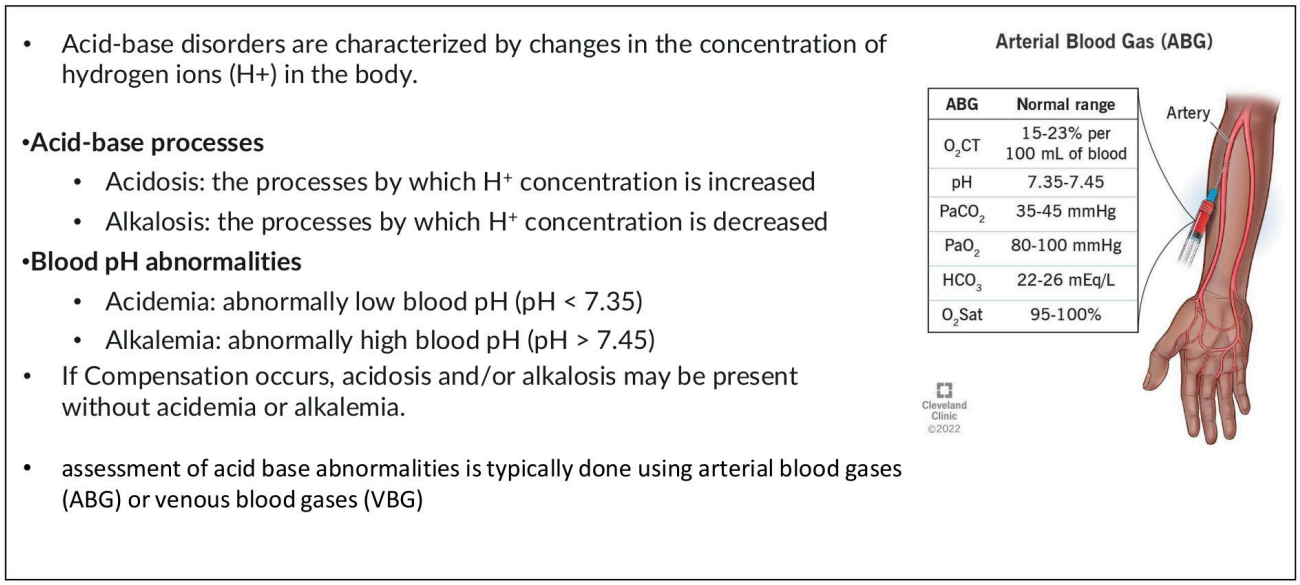
- ECF volume is set by **Na^+ content** (filtration/reabsorption)
- **ANP** (atria, volume overload) opposes Na^+ reabsorption
- osmolarity is controlled by **ADH (vasopressin)**, released with osmolarity changes of **< 1 %**

2.2 pH & Buffers (Slides 8–9) ⭐⭐

- pH reflects **$[H^+]$**; balance is kept by **buffers + lungs + kidneys**.
- ⭐ **HCO_3^- is the most important buffer**:
  1. it is **consumed by metabolic acid**, making **CO_2 that is exhaled**
  2. it can be **regenerated by the kidney**
- **$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$**: lungs control CO_2, kidneys control HCO_3^-.

Term	Meaning
Acidosis	**Process** raising $[H^+]$
Alkalosis	**Process** lowering $[H^+]$
⭐ **Acidaemia**	**Blood pH < 7.35**
⭐ **Alkalaemia**	**Blood pH > 7.45**

⭐ With compensation (or a mixed disorder), **acidosis/alkalosis can exist without acidaemia/alkalaemia**. Assessment uses an **ABG** (or VBG).



Slide 9: Normal ABG ranges

Normal ABG (slide 9):

Parameter	Normal
pH	**7.35–7.45**
$PaCO_2$	**35–45 mmHg**
PaO_2	**80–100 mmHg**
HCO_3^-	**22–26 mEq/L**
O_2 content	15–23 mL/100 mL
O_2 saturation	95–100 %

2.3 Respiratory Acidosis (Slides 10–11) ⭐⭐

Basics:

- **Definition: ↑ $PaCO_2$, ↓ pH**
- **Mechanism: alveolar hypoventilation** → CO_2 retention

Causes:

Level	Causes
Central (CNS)	**Drugs** (*[added]* opioids, benzodiazepines), **stroke**, trauma, haemorrhage
Airway	**Obstruction, asthma**
Parenchyma	**Emphysema**, pneumoconiosis, **bronchitis**, ARDS, barotrauma, pneumonia
Neuromuscular	Poliomyelitis, **kyphoscoliosis**, **myasthenia**, muscular dystrophies
Miscellaneous	**Obesity hypoventilation**, **ALS**

Clinical features (slide 11): somnolence, **myoclonus with asterixis** (CO_2 flap), **headache** (cerebral vasodilatation); the picture of respiratory failure.

Compensation (renal, takes days):

- the kidney **↑ HCO_3^- reabsorption (PCT)** and **↑ H^+ excretion (DCT)**

	Rule
⭐ **Acute**	**HCO_3^- ↑ 1 per 10 mmHg ↑ $PaCO_2$** → expected $HCO_3^- = 24 + 0.1 \times (PaCO_2 - 40)$
⭐ **Chronic**	**HCO_3^- ↑ 3.5 per 10 mmHg ↑ $PaCO_2$** → expected $HCO_3^- = 24 + 0.35 \times (PaCO_2 - 40)$

Treatment:

- **treat the cause**
- **O_2 if PaO_2 is low**, but ⚠️ **careful in COPD** (*[added]* target SpO_2 88–92 %)
- **intubation and mechanical ventilation** for **severe acidosis (pH < 7)**, **mental deterioration**, or **impending respiratory fatigue** (*[added]* **NIV/BiPAP** first in COPD exacerbation with pH 7.25–7.35)

2.4 Respiratory Alkalosis (Slides 12–13) ⭐⭐

Basics:

- **Definition: ↓ $PaCO_2$, ↑ pH**
- **Mechanism:** ↑ RR and/or tidal volume → **alveolar hyperventilation** → CO_2 washout

Causes:

Group	Causes
CNS stimulation	**Pain, anxiety**, psychosis, **fever**, CVA, meningitis, encephalitis, tumour, trauma
Hypoxaemia	**High altitude**, pneumonia, pulmonary oedema, **severe anaemia**, ⭐ **PE**
Drugs/hormones	⭐ **Pregnancy** (progesterone), ⭐ **salicylates (early)**
Miscellaneous	⭐ **Sepsis** (early), **hepatic failure**, mechanical over-ventilation, heat exposure

Clinical features (slide 13):

- light-headedness, dizziness, anxiety
- ⭐ **paraesthesia and perioral numbness** (*[added]* ↓ ionised Ca^{2+} → tetany, carpopedal spasm)

- confusion, chest discomfort

Compensation:

	Rule
⭐ **Acute**	**HCO_3^- ↓ 2 per 10 mmHg ↓ $PaCO_2$** → expected $HCO_3^- = 24 - 0.2 \times (40 - PaCO_2)$
⭐ **Chronic**	**HCO_3^- ↓ 4 per 10 mmHg ↓ $PaCO_2$** → expected $HCO_3^- = 24 - 0.4 \times (40 - PaCO_2)$

[added] Some texts use 5 for chronic alkalosis; learn the slide's **4**.

Treatment:

- **treat the cause**
- sometimes **no treatment needed** (e.g. **pregnancy**)
- **rebreathing into a paper bag** (anxiety)

> ⚠️ *[added]* **Paper-bag rebreathing** is **discouraged** now: it can cause **hypoxaemia**, and it can mask hyperventilation from **PE, MI, asthma or DKA**. Exclude organic causes first; coached slow breathing is safer.
>
> 🔁 **Recap: Respiratory disorders.**
>
> - **Respiratory acidosis:** ↑ CO_2 from hypoventilation (CNS, airway, parenchyma, neuromuscular, obesity) → somnolence, asterixis. HCO_3^- ↑ **1 (acute) / 3.5 (chronic)** per 10. Treat the cause, cautious O_2, ventilate if pH < 7.
> - **Respiratory alkalosis:** ↓ CO_2 from hyperventilation (anxiety, pain, hypoxaemia/PE, pregnancy, early salicylate, sepsis, liver failure) → paraesthesia. HCO_3^- ↓ **2 (acute) / 4 (chronic)** per 10.
>
> 🔗 **Link:** the metabolic disorders flip the logic: **HCO_3^- moves first**, and the lungs compensate **immediately**.
>
> ▶️ **Up next: Metabolic acidosis.**

2.5 Metabolic Acidosis: Causes & Anion Gap (Slides 14–15) ⭐⭐⭐

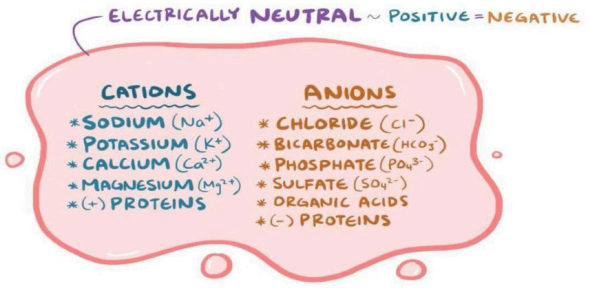
Basics:

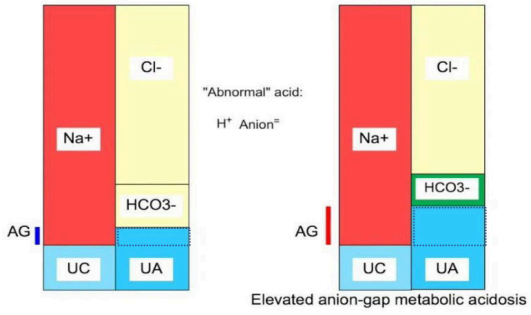
- **Definition: ↓ HCO_3^-, ↓ pH**
- **Mechanism: ↑ production/ingestion of H^+** (gap) or **loss of HCO_3^-** (non-gap)

Anion gap (AG) (slide 15):

- AG = difference between the main measured **cations** and **anions**
- ⭐ **$AG = Na^+ - (Cl^- + HCO_3^-)$**; **normal 6–12 mmol/L**
- alternative (with K^+): **$(Na^+ + K^+) - (Cl^- + HCO_3^-)$**; **normal 10–16**
- the summary flowchart (slide 24) uses **> 12 = high**, **8–12 = normal**

- The anion gap (AG) is the difference between primary measured cations and the primary measured anions
- The AG is calculated as follows: **$AG = Na^+ – (Cl^- + HCO3^-)$**
- The “normal” value for the AG ranges from 6 to 12 mmol/L
- Another Formula: $([Na^+] + [K^+]) - ([Cl^-] + [HCO_3^-])$, with a normal range of 10–16 mEq/L.





Slide 15: Anion gap: unmeasured anions; normal vs elevated AG

[added] **Albumin correction:** albumin is the main unmeasured anion. **Corrected AG = AG + 2.5 × (4 − albumin g/dL)**. Hypoalbuminaemia (sepsis, cirrhosis, nephrotic syndrome) can **hide** a high-gap acidosis.

⭐ **High AG: "MUDPILES"**	⭐ **Normal AG (hyperchloraemic): "HARDASS"**
Methanol (formic acid)	**H**yperalimentation (TPN; slide 24 adds hyperchloraemia)
Uraemia	**A**cetazolamide
DKA (*[added]* + alcoholic and starvation ketoacidosis)	**R**enal tubular acidosis
Propylene glycol	**D**iarrhoea (⭐ commonest)
INH / **I**ron	**A**ddison's disease
Lactic acidosis	**S**pironolactone
Ethanol / **E**thylene glycol (oxalic acid)	**S**aline infusion (large volumes)
Salicylates (**late**)	

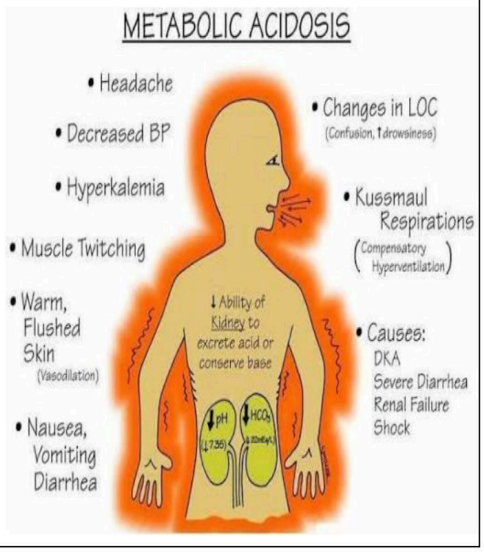
[added] Modern alternative for the high-AG list, **"GOLD MARK"**: **G**lycols, **O**xoproline (paracetamol), **L**-lactate, **D**-lactate, **M**ethanol, **A**spirin, **R**enal failure, **K**etoacidosis.

2.6 Metabolic Acidosis: Clinical, Compensation, Treatment (Slide 16) ⭐⭐

Clinical Features:
- Hyperventilation (Kussmaul respiration)
- Decreased in Cardiac output and tissue perfusion

Compensation:
- Hyperventilation → decrease $PaCO_2$ immediately
- **Winter's formula:** predicted $PaCO_2 = (1.5 \times [HCO_3] + 8) \pm 2$

Treatment:
- Treat the underlying cause
- Sodium bicarbonate is sometimes used in severe acidosis
- Mechanical ventilation might be needed if the patient is fatigued
- For ethylene glycol and methanol ingestion, treat with fomepizole and hemodialysis



Slide 16: Metabolic acidosis features

Clinical features:

- ⭐ **Kussmaul respiration** (deep, rapid)
- **↓ cardiac output and tissue perfusion** (*[added]* severe acidaemia depresses myocardial contractility and vasopressor response)
- *[added]* hyperkalaemia (H^+/K^+ shift)

Compensation:

- **hyperventilation → ↓ $PaCO_2$ immediately**
- ⭐ **Winter's formula: expected $PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$**
  - actual $PaCO_2$ **higher** than expected → **coexisting respiratory acidosis** (e.g. fatigue)
  - actual $PaCO_2$ **lower** → **coexisting respiratory alkalosis** (e.g. salicylates, sepsis)
  - *[added]* quick check: expected $PaCO_2$ ≈ the last two digits of the pH (pH 7.25 → ~25)

Treatment:

- **treat the cause** (*[added]* DKA: fluids + insulin + K^+; lactic: perfusion)
- **sodium bicarbonate** sometimes in **severe acidosis** (*[added]* usually pH < 7.1–7.2, especially with AKI or bicarbonate-loss causes; little benefit in DKA)
- **mechanical ventilation** if the patient is fatiguing
- ⭐ **ethylene glycol and methanol → fomepizole + haemodialysis**

[added] **Toxic alcohols:**

- **fomepizole** blocks **alcohol dehydrogenase**, stopping formation of the toxic acids (formic acid → blindness; glycolic/oxalic acid → AKI, calcium oxalate crystals)
- an **osmolar gap > 10** suggests unmetabolised alcohol
- ethanol is the alternative ADH blocker

2.7 Metabolic Alkalosis (Slides 17–19) ⭐⭐⭐

Basics:

- **Definition: ↑ HCO_3^-, ↑ pH**
- **Mechanism: loss of H^+** or **↑ HCO_3^- production/ingestion**

	⭐ **Chloride-responsive (urine Cl^- < 25 mEq/L)**	⭐ **Chloride-resistant (urine Cl^- > 40 mEq/L)**
Causes (slide 17)	**Hypovolaemia** (contraction alkalosis); **renal losses (diuretics)**; **cystic fibrosis**; dietary chloride deficiency; *[added]* ⭐ **vomiting / NG suction** (commonest)	**Loop or thiazide diuretics**; ⭐ **hyperaldosteronism**; **Cushing syndrome**; **high alkali load** (antacids, alkalinisation therapy)
Volume *[added]*	**Depleted**	Often **normal/expanded** + **hypertension** (mineralocorticoid excess)
Treatment (slide 19)	⭐ **Isotonic saline** (↑ urinary HCO_3^- excretion, corrects ECF loss)	Slide: consider **bicarbonate excess** → **acetazolamide**

⚠️ **Slide inconsistencies:**

- **Diuretics appear in both columns.** *[added]* Resolution: **after** diuretics have worn off, urine Cl^- is **low** (responsive); **during** active use, urine Cl^- is **high** (resistant).
- **Vomiting**, the commonest cause and the subject of **Case 3**, is mentioned on slide 18 but **missing from the slide 17 cause table**.
- **Urine Cl^- 25–40 mEq/L** is left undefined by the slide's cut-offs.
- **Slide 19:** acetazolamide is used mainly for alkalosis with **volume overload** (e.g. post-diuresis in HF). For **chloride-resistant** alkalosis from **mineralocorticoid excess**, treat the cause *[added]*: **spironolactone/eplerenone**, adrenalectomy, **K^+ repletion**.

Clinical (slide 18):

- **no characteristic signs**
- **muscle weakness, cramps, spasms, tremor**, fatigue, irritability, **confusion, seizures** (*[added]* ↓ ionised Ca^{2+}, **hypokalaemia**, arrhythmias)
- ⭐ the **history** is most helpful: **vomiting, gastric drainage, diuretic therapy**

Compensation:

- **hypoventilation → ↑ $PaCO_2$ immediately**
- ⭐ **expected $PaCO_2 = 0.7 \times (HCO_3^- - 24) + 40 \pm 2$**
- *[added]* compensation is limited by hypoxaemia; $PaCO_2$ rarely exceeds ~55

⭐ **The first step once metabolic alkalosis is diagnosed: measure urinary chloride.**

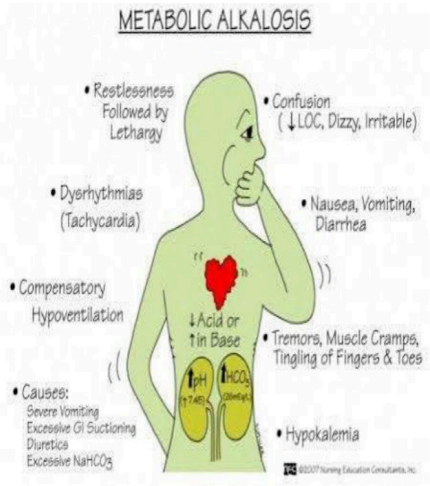
Clinical Features:

- There are no characteristic signs and symptoms
- Muscle weakness, cramps, spasms, tremors, fatigue, irritability, and changes in mental status like confusion or seizures
- The patient's medical history is most helpful (look for vomiting, gastric drainage, diuretic therapy, and so on)

Compensation:

- Hypoventilation → increases $PaCO_2$ immediately
- $PaCO_2$ = [0.7 x (HCO_3 – 24)] + 40 ± 2

The first step if metabolic alkalosis is diagnosed is to measure urinary chloride



Slide 18: Metabolic alkalosis features

🔁 **Recap: Metabolic disorders.**

- **Metabolic acidosis:** ↓ HCO_3^-. Calculate **AG = Na − (Cl + HCO_3)**: high (MUDPILES) vs normal (HARDASS). **Winter's: 1.5 × HCO_3 + 8 ± 2.** Kussmaul breathing. Treat the cause; HCO_3^- if severe; **fomepizole + dialysis** for methanol/ethylene glycol.
- **Metabolic alkalosis:** ↑ HCO_3^-. **Urine Cl^- first**: < 25 responsive (vomiting, volume loss → **saline**) vs > 40 resistant (aldosterone, Cushing's, alkali, active diuretics). **$PaCO_2$ = 0.7 × (HCO_3 − 24) + 40 ± 2.**

🔗 **Link:** all four disorders are pulled into one systematic routine.

▶️ **Up next: Patient approach.**

2.8 Patient Approach (Slides 20–23) ⭐⭐⭐

⭐ **pH and $PaCO_2$ are the most important values** (slide 20).

Step 1: pH:

pH	Meaning
< 7.35	Acidaemia: primary **acidosis**
> 7.45	Alkalaemia: primary **alkalosis**
7.35–7.45	**Normal, or mixed** acidosis + alkalosis

Step 2: $PaCO_2$ (35–45) ⭐ SMORE:

- **S**ame direction as pH → **M**etabolic
- **O**pposite direction → **RE**spiratory

pH	$PaCO_2$	Disorder
↓	↑	**Respiratory acidosis**
↑	↓	**Respiratory alkalosis**
↓	↓	**Metabolic acidosis**
↑	↑	**Metabolic alkalosis**

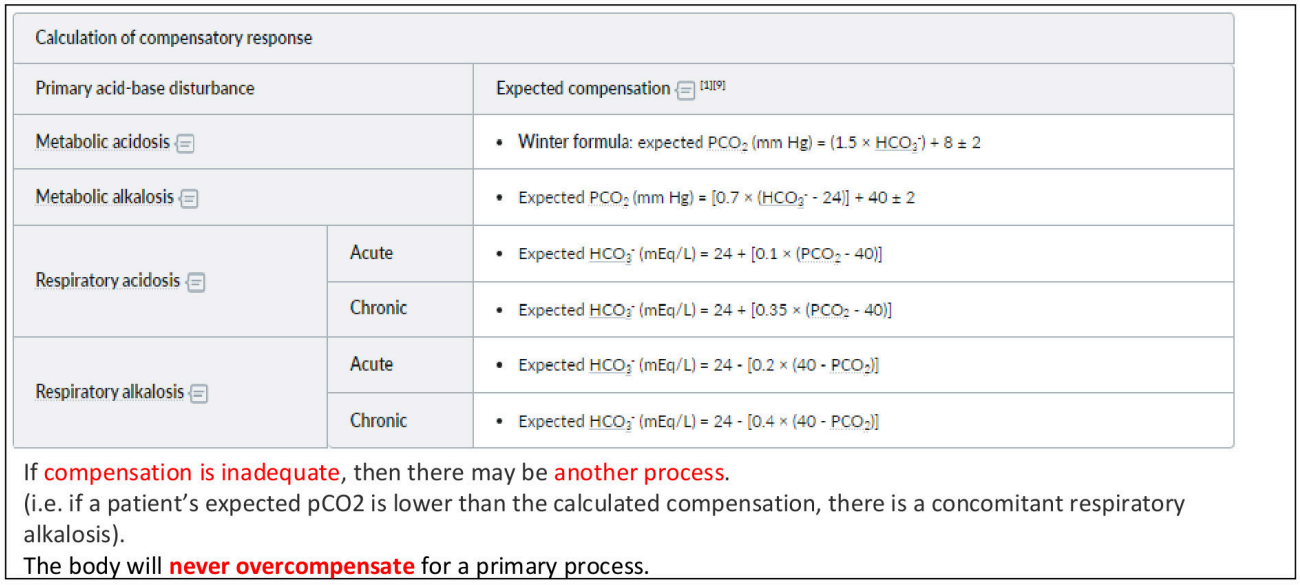
Step 3: HCO_3^- (22–26):

HCO_3^-	Means
High	Metabolic alkalosis **or** compensated respiratory acidosis
Normal	**Uncompensated** respiratory disorder
Low	Metabolic acidosis **or** compensated respiratory alkalosis

Step 4: PaO_2: high = hyperoxaemia; low = hypoxaemia.

Step 5: chronicity and compensation using the formulas (table below).

Step 6: ⭐ calculate the AG. ⭐ Check it even without acidaemia: a metabolic acidosis may be hiding.



Calculation of compensatory response		
Primary acid-base disturbance		Expected compensation [1][9]
Metabolic acidosis		• Winter formula: expected PCO_2 (mm Hg) = (1.5 × HCO_3^-) + 8 ± 2
Metabolic alkalosis		• Expected PCO_2 (mm Hg) = [0.7 × (HCO_3^- - 24)] + 40 ± 2
Respiratory acidosis	Acute	• Expected HCO_3^- (mEq/L) = 24 + [0.1 × (PCO_2 - 40)]
	Chronic	• Expected HCO_3^- (mEq/L) = 24 + [0.35 × (PCO_2 - 40)]
Respiratory alkalosis	Acute	• Expected HCO_3^- (mEq/L) = 24 - [0.2 × (40 - PCO_2)]
	Chronic	• Expected HCO_3^- (mEq/L) = 24 - [0.4 × (40 - PCO_2)]

If compensation is inadequate, then there may be another process.
(i.e. if a patient's expected pCO2 is lower than the calculated compensation, there is a concomitant respiratory alkalosis).
The body will never overcompensate for a primary process.

Slide 23: Expected compensation table

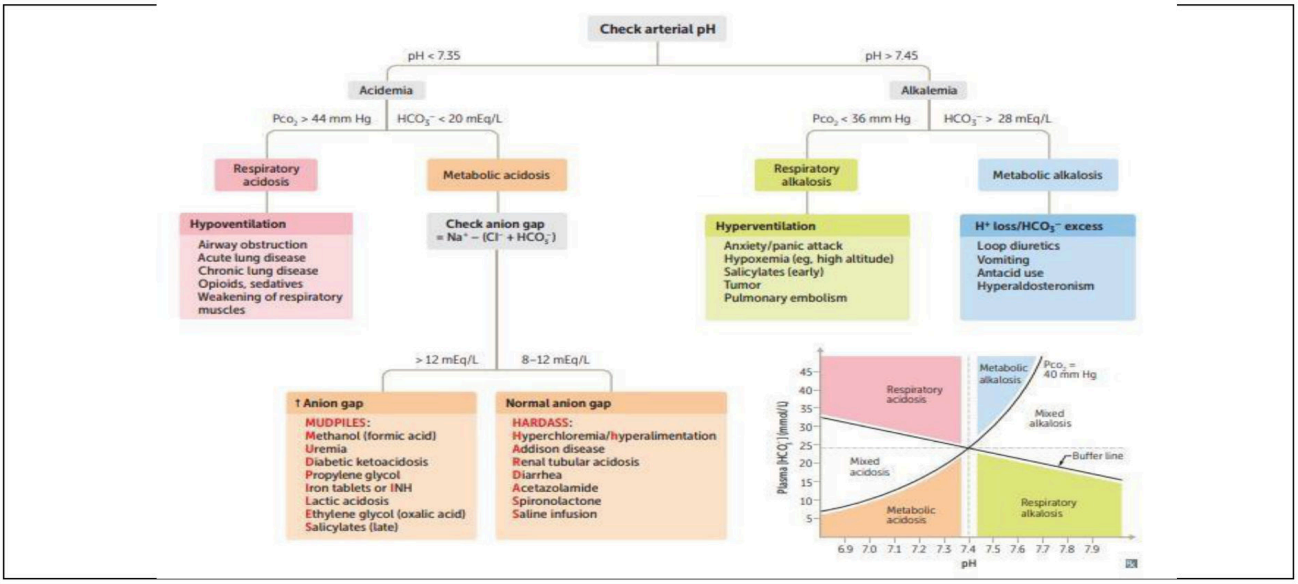
⭐ Primary disorder	Expected compensation (slide 23)
Metabolic acidosis	**$PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$** (Winter's)
Metabolic alkalosis	**$PaCO_2 = 0.7 \times (HCO_3^- - 24) + 40 \pm 2$**
Respiratory acidosis, acute	**$HCO_3^- = 24 + 0.1 \times (PaCO_2 - 40)$**
Respiratory acidosis, chronic	**$HCO_3^- = 24 + 0.35 \times (PaCO_2 - 40)$**
Respiratory alkalosis, acute	**$HCO_3^- = 24 - 0.2 \times (40 - PaCO_2)$**
Respiratory alkalosis, chronic	**$HCO_3^- = 24 - 0.4 \times (40 - PaCO_2)$**

Mnemonic for the respiratory multipliers, "1-3.5 up, 2-4 down": acidosis up by **1 / 3.5**; alkalosis down by **2 / 4** (acute / chronic, per 10 mmHg).

Interpreting compensation:

- ⭐ **inadequate compensation → another process.** E.g. if $PaCO_2$ is **lower** than the expected value → **concomitant respiratory alkalosis**.
- ⭐ **The body never overcompensates** for a primary process. A pH on the "other side" of 7.40 means a second disorder.

2.9 Summary Flowchart (Slide 24) ⭐



Slide 24: Summary acid–base flowchart

Flowchart contents:

- **Acidaemia (pH < 7.35):**
  - **$PaCO_2$ > 44** → respiratory acidosis (hypoventilation: airway obstruction, acute/chronic lung disease, opioids/sedatives, weak respiratory muscles)
  - **HCO_3^- < 20** → metabolic acidosis → **AG** (> 12 → MUDPILES; 8–12 → HARDASS)
- **Alkalaemia (pH > 7.45):**
  - **$PaCO_2$ < 36** → respiratory alkalosis (hyperventilation: anxiety/panic, hypoxaemia/high altitude, early salicylates, tumour, PE)
  - **HCO_3^- > 28** → metabolic alkalosis (**H^+ loss / HCO_3^- excess**: loop diuretics, vomiting, antacids, hyperaldosteronism)

⚠️ **Threshold inconsistency:** the flowchart uses $PaCO_2$ > 44 / < 36 and HCO_3^- < 20 / > 28. The text slides use **35–45** and **22–26**. Use the text ranges for normal values; the flowchart cut-offs are pragmatic "clearly abnormal" values.

Tic-Tac-Toe method (named in objectives, not taught on slides) [added]

	ACID	NORMAL	BASE
pH	< 7.35 **7.20**	7.35–7.45	> 7.45
$PaCO_2$	> 45	35–45	< 35 **25**
HCO_3^-	< 22 **10**	22–26	> 26

Note: high CO_2 = ACID column; low CO_2 = BASE column

Example = slide Case 4: pH 7.20 · $PaCO_2$ 25 · HCO_3^- 10

Read the grid

1. Circle each value in its column.
2. The value sharing pH's column
= PRIMARY disorder
(CO_2 → respiratory; HCO_3 → metabolic).
3. Other value in the OPPOSITE column
= compensation.
4. Other value NORMAL = uncompensated.
5. pH normal + both abnormal
= fully compensated (or mixed).
6. Both in pH's column = MIXED disorder.

Full 6-step ABG routine (slides 20–23 + added steps)

1 pH → acidaemia / alkalaemia / normal

2 $PaCO_2$ vs pH: SAME direction → Metabolic; OPPOSITE → REspiratory (SMORE)

3 Compensation formula → appropriate? if not → second process (body never overcompensates)

4 Anion gap = Na − (Cl + HCO_3): normal 6–12 (8–12 on summary slide) — check it EVEN if pH normal

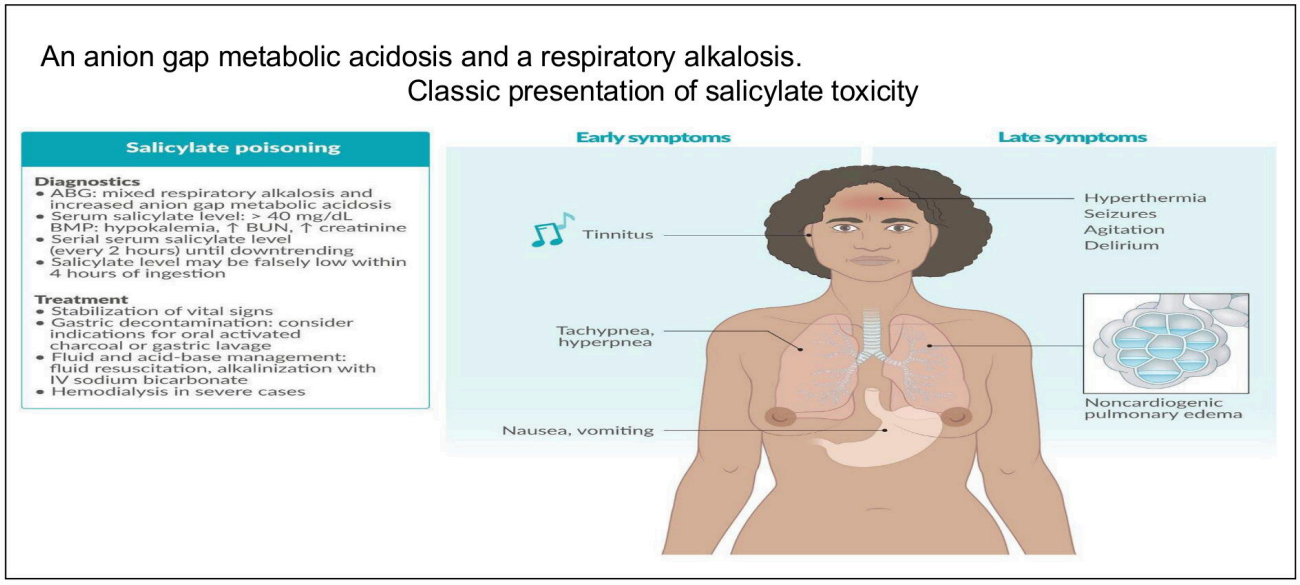
5 High AG → delta ratio ΔAG ÷ ΔHCO_3: < 1 + NAGMA, 1–2 pure, > 2 + metabolic alkalosis [added]

6 Suspect toxic alcohol → osmolar gap; correct AG for albumin (+2.5 per 1 g/dL ↓) [added]

Case 4 worked: AG = 135 − 105 = 30 (high) · Winter = 1.5×10 + 8 = 23 ± 2 → 25 OK · Δratio = 18/14 ≈ 1.3 → pure HAGMA

Added diagram: Tic-Tac-Toe grid and the full ABG routine (worked on Case 4)

2.10 Salicylate Toxicity (Case 1 Answer, Slide 26) ⭐



Slide 26: Salicylate poisoning

Feature	Detail
ABG	⭐ **Mixed respiratory alkalosis (early, direct respiratory-centre stimulation) + high-AG metabolic acidosis (late)**
Level	Serum salicylate > 40 mg/dL; **repeat every 2 h until falling** (it may be falsely low within 4 h of ingestion)
Other labs	**Hypokalaemia**, ↑ BUN, ↑ creatinine
Early symptoms	⭐ **Tinnitus**, tachypnoea/hyperpnoea, **nausea, vomiting**

Feature	Detail
Late symptoms	**Hyperthermia, seizures, agitation, delirium**; **non-cardiogenic pulmonary oedema**
Treatment	Stabilise; **activated charcoal** or gastric lavage; fluids; ⭐ **alkalinisation with IV sodium bicarbonate** (urine + serum); ⭐ **haemodialysis** in severe cases

2.11 Cases (Slides 25–32): Verified ⭐⭐⭐

Case	Data	Slide key	My check
1	pH 7.24, $PaCO_2$ 22, PaO_2 92; Na 138, Cl 102, HCO_3^- 15	**HAGMA + respiratory alkalosis: salicylate toxicity**	✅ AG = 138 − 117 = **21** (high). Winter's = 1.5 × 15 + 8 = **30.5 ± 2**; actual **22 < 28.5** → extra **respiratory alkalosis**
2	COPD smoker; pH 7.33, $PaCO_2$ 55, HCO_3^- 30; Na 140, Cl 100	**Chronic compensated respiratory acidosis**	✅ Chronic expected HCO_3^- = 24 + 0.35 × 15 = **29.25** ≈ 30 (acute would be 25.5). AG = 10 (normal). *[added]* "Compensated" here means **appropriately (partially) compensated**: pH is still < 7.35
3	3 days of vomiting; pH 7.50, $PaCO_2$ 48, HCO_3^- 34; Na 138, Cl 92, K 3.2	❌ **E. "Acute uncompensated metabolic alkalosis"**	⚠️ **The key contradicts the slide's own formula.** Expected $PaCO_2$ = 0.7 × (34 − 24) + 40 = **47 ± 2 (45–49)**; actual **48** → **appropriate respiratory compensation**. "Uncompensated" is wrong. See below
4	Sepsis, lactate 5; pH 7.20, $PaCO_2$ 25, HCO_3^- 10; Na 135, Cl 95	**A. HAGMA with appropriate respiratory compensation**	✅ AG = 135 − 105 = **30**. Winter's = 1.5 × 10 + 8 = **23 ± 2 (21–25)**; actual 25 → appropriate. *[added]* Delta ratio = (30 − 12)/(24 − 10) = 18/14 ≈ **1.3** → pure HAGMA (lactic)

⚠️ **Case 3 in detail** *[analysis added]*:

- The correct description is **metabolic alkalosis (chloride-responsive, from vomiting: H^+ and Cl^- loss, volume contraction, hypokalaemia) with appropriate respiratory compensation**.
- The "acute vs chronic" label in the options is **not meaningful** for metabolic disorders. Respiratory compensation is **immediate** (slide 18), and the compensation formula has no acute/chronic forms.
- Among the options, the closest correct answers are **D or H ("compensated metabolic alkalosis")**, not E.
- If the lecturer insists on E, they are probably using "uncompensated" to mean "pH still abnormal" (i.e. partially compensated). That conflicts with the formula taught on slides 18 and 23.
- **Management: urine Cl^- (expect < 25) → IV 0.9 % saline + KCl**.
- **Ask the lecturer** which wording they will accept.

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3. High-Yield Comparison Tables

Table A: The Four Primary Disorders

	Resp. acidosis	Resp. alkalosis	Met. acidosis	Met. alkalosis
Primary change	↑ **$PaCO_2$**	↓ **$PaCO_2$**	↓ **HCO_3^-**	↑ **HCO_3^-**

	Resp. acidosis	Resp. alkalosis	Met. acidosis	Met. alkalosis
pH	↓	↑	↓	↑
Compensation	↑ HCO_3^- (renal, **days**)	↓ HCO_3^- (renal)	↓ $PaCO_2$ (**immediate**)	↑ $PaCO_2$ (**immediate**)
Formula	**+1 / +3.5** per 10	**−2 / −4** per 10	**1.5 × HCO_3 + 8 ± 2**	**0.7 × (HCO_3 − 24) + 40 ± 2**
Clinical	Somnolence, asterixis, headache	Paraesthesia, perioral numbness, dizziness	**Kussmaul**, ↓ CO	Weakness, cramps, tremor, seizures
Key causes	COPD, sedatives, neuromuscular, obesity	Anxiety, PE, pregnancy, early salicylate, sepsis, liver failure	MUDPILES / HARDASS	Vomiting, diuretics, hyperaldosteronism, Cushing's, alkali
Key treatment	Cause, cautious O_2, **ventilate**	Cause (± none in pregnancy)	Cause, **$NaHCO_3$ if severe**, **fomepizole/HD**	**Saline** (Cl-responsive); treat cause (Cl-resistant)

Table B: High AG vs Normal AG Metabolic Acidosis

	High AG (> 12)	Normal AG (6/8–12)
Mechanism	**Added acid** (unmeasured anion)	**HCO_3^- loss**, replaced by **Cl^-** (hyperchloraemic)
Mnemonic	**MUDPILES**	**HARDASS**
Classic	DKA, lactic, renal failure, toxic alcohols, late salicylate	**Diarrhoea**, RTA, acetazolamide, saline, Addison's
[added] Next test	Lactate, ketones, creatinine, **osmolar gap**, salicylate level	**Urine AG** (negative = GI loss; positive = RTA)

Table C: Chloride-Responsive vs Chloride-Resistant Metabolic Alkalosis

	Responsive	Resistant
Urine Cl^-	**< 25 mEq/L**	**> 40 mEq/L**
Causes	**Vomiting/NG suction**, **remote diuretics**, hypovolaemia, CF, low-Cl diet	**Active diuretics**, **hyperaldosteronism**, **Cushing's**, **alkali load** (*[added]* Bartter/Gitelman, severe hypokalaemia)
Volume / BP *[added]*	Low volume	Normal/high, often **HTN**
Treatment	**0.9 % saline** (+ K^+)	Slide: **acetazolamide**; *[added]* treat the cause, **spironolactone**, K^+

Table D: Compensation Speed

Primary	Compensating organ	Speed
Metabolic	**Lungs**	**Minutes–hours** ("immediately")
Respiratory	**Kidneys**	**3–5 days** for full (chronic) compensation *[added]*

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Acidosis = acidaemia	❌ With compensation or a **mixed** disorder, pH can be normal
Normal pH = no disorder	❌ Check $PaCO_2$, HCO_3^- and **AG**: "**a metabolic acidosis may be hiding**"
Compensation returns pH to normal	❌ **Never overcompensates**; usually **partial**; pH on the opposite side → second disorder
$PaCO_2$ lower than Winter's	**+ respiratory alkalosis** (salicylate, sepsis)
$PaCO_2$ higher than Winter's	**+ respiratory acidosis** (fatigue, sedation)
Salicylate = metabolic acidosis	**Early: respiratory alkalosis**; **late: HAGMA** → classic **mixed** picture
Diuretics → Cl-resistant	Only **while acting**; afterwards urine Cl^- falls → **responsive**
Give O_2 for hypoxaemia	⚠️ **COPD/CO_2 retainers**: controlled O_2 (88–92 %)
Bicarbonate fixes acidosis	Only for **severe** cases (slide "sometimes"); treat the cause
Paper bag for hyperventilation	*[added]* ⚠️ Risky; exclude PE/MI/DKA
AG normal 6–12	With K^+ 10–16; summary slide 8–12; *[added]* correct for **albumin**

Pattern recognition:

- **tinnitus + tachypnoea + HAGMA with low $PaCO_2$ → salicylate**
- **visual loss + HAGMA + osmolar gap → methanol**
- **AKI + oxalate crystals + HAGMA → ethylene glycol**
- **diarrhoea + normal AG + ↓ K^+** → GI HCO_3^- loss
- **vomiting + ↓ Cl^- + ↓ K^+ + ↑ HCO_3^- → Cl-responsive alkalosis**
- **HTN + ↓ K^+ + ↑ HCO_3^- + urine Cl^- high → hyperaldosteronism**
- **COPD + ↑ $PaCO_2$ + HCO_3^- ≈ 24 + 0.35 × $\Delta PaCO_2$ → chronic respiratory acidosis**

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5. Clinical Correlations

- **COPD exacerbation:**
  - acute-on-chronic respiratory acidosis: HCO_3^- falls **between** the acute and chronic expected values *[added]*
  - **controlled O_2**; **NIV** at pH 7.25–7.35; intubate if worse
- **DKA** *[added]*:
  - HAGMA with Kussmaul breathing and ketones
  - treat with fluids, insulin and **K^+** (total-body K^+ is depleted even if the serum level is high)
  - bicarbonate only if pH < 6.9
- **Sepsis:**
  - early **respiratory alkalosis** → later **lactic acidosis** (Case 4)
  - lactate drives resuscitation
- **Toxic alcohols** (methanol, ethylene glycol):
  - **fomepizole + haemodialysis** (slide)
  - *[added]* calculate the **osmolar gap**
- **Salicylate overdose: urinary alkalinisation with $NaHCO_3$**, serial levels, **haemodialysis** if severe. Never intubate without matching the high minute ventilation *[added]*.
- **Persistent vomiting / NG suction:**

- **urine Cl^-** first
- **normal saline + KCl**
- *[added]* paradoxical aciduria occurs in severe volume and K^+ depletion

- **Hyperaldosteronism / Cushing's** (Adrenal lectures): **HTN + hypokalaemia + metabolic alkalosis**, saline-unresponsive.

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6. Rapid Revision (Last-Day)

- **Kidney anatomy:** retroperitoneal, **T12–L3**, right lower; nephron = glomerulus → PCT → loop of Henle → DCT → collecting duct.
- **Kidney functions:**
  - excretory; regulatory (volume, pH)
  - **endocrine** (EPO, renin, angiotensin, prostaglandins)
  - **metabolic** (vitamin D, insulin breakdown)
  - ANP opposes Na^+ reabsorption; **ADH** responds to < 1 % osmolarity change
- **Buffers: HCO_3^- is the key buffer**: consumed by acid → CO_2 exhaled; **regenerated by the kidney**.
- **Definitions: acidaemia pH < 7.35; alkalaemia > 7.45**; acidosis/alkalosis = processes.
- **Normal ABG: pH 7.35–7.45**, **$PaCO_2$ 35–45**, **PaO_2 80–100**, **HCO_3^- 22–26**.
- **Respiratory acidosis:**
  - ↑ $PaCO_2$ from hypoventilation (CNS drugs/stroke, airway/asthma, COPD/ARDS/pneumonia, neuromuscular, obesity, ALS)
  - **somnolence, asterixis, headache**
  - HCO_3^- **+1 acute / +3.5 chronic** per 10
  - treat the cause, careful O_2 in COPD, **ventilate if pH < 7, mental decline or fatigue**
- **Respiratory alkalosis:**
  - ↓ $PaCO_2$ (pain, anxiety, fever, CNS, hypoxaemia, altitude, anaemia, **PE**, **pregnancy**, **early salicylate**, **sepsis**, **liver failure**)
  - **paraesthesia, perioral numbness**
  - HCO_3^- **−2 acute / −4 chronic** per 10
  - treat the cause; none in pregnancy; paper bag (slide; outdated)
- **Metabolic acidosis:**
  - **$AG = Na - (Cl + HCO_3)$**, normal **6–12** (with K^+ 10–16)
  - **MUDPILES** vs **HARDASS**
  - **Kussmaul**, ↓ CO
  - **Winter's: $PaCO_2 = 1.5 \times HCO_3 + 8 \pm 2$**
  - treat cause ± **$NaHCO_3$**, ventilate if tired, **fomepizole + HD** for methanol/ethylene glycol
- **Metabolic alkalosis:**
  - **urine Cl^- first**: **< 25 responsive** (hypovolaemia, diuretics, CF, low Cl^-, *[added]* vomiting) → **saline**
  - **> 40 resistant** (diuretics, **hyperaldosteronism, Cushing's, alkali load**) → slide: acetazolamide
  - compensation: **$PaCO_2 = 0.7 \times (HCO_3 - 24) + 40 \pm 2$**
  - weakness, cramps, tremor, confusion, seizures; history (vomiting, drainage, diuretics) is key
- **Approach:**
  - **pH → $PaCO_2$ (SMORE) → HCO_3^- → PaO_2 → compensation → AG**
  - **check AG even if pH is normal**

- **never overcompensates**; inadequate compensation = second process

- **Salicylate toxicity: resp alkalosis + HAGMA**, **tinnitus**, N/V, hyperthermia, seizures, non-cardiogenic pulmonary oedema; **charcoal, bicarbonate alkalinisation, HD**.
- **Case keys:**
  1. salicylate (HAGMA + resp alkalosis)
  2. chronic compensated respiratory acidosis
  3. slide says **E** but the formula shows **appropriately compensated metabolic alkalosis**
  4. **HAGMA with appropriate compensation**

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7. Exam-Style Questions

Q1. (Slide Case 1) ABG: pH 7.24, $PaCO_2$ 22, PaO_2 92; Na 138, Cl 102, HCO_3^- 15. Most likely cause?

> **Answer: High-AG metabolic acidosis + respiratory alkalosis = salicylate toxicity** (slide 26).
> - AG = 21. Expected $PaCO_2$ (Winter's) = 30.5 ± 2; actual 22 is **too low** → additional respiratory alkalosis.

Q2. (Slide Case 2) A 68-year-old COPD smoker with worsening dyspnoea: pH 7.33, $PaCO_2$ 55, HCO_3^- 30; Na 140, Cl 100. Which describes the acid–base state? **A. Chronic compensated respiratory acidosis** · B. Acute compensated respiratory acidosis · C. HAGMA · D. Normal-AG metabolic acidosis

> **Answer: A** (slide key, verified).
> - Chronic expected HCO_3^- = 24 + 0.35 × 15 ≈ **29**; acute would be ~25.5.
> - AG = 10, so no metabolic acidosis.

Q3. (Slide Case 3) A 35-year-old man with 3 days of vomiting: pH 7.50, $PaCO_2$ 48, HCO_3^- 34; Na 138, Cl 92, K 3.2. Best description and first diagnostic step?

> **Slide key: E (acute uncompensated metabolic alkalosis).**
>
> **My answer** *[reasoning added; key disputed]*: **metabolic alkalosis with appropriate respiratory compensation** (expected $PaCO_2$ 47 ± 2; actual 48).
> - This is the **chloride-responsive** type from gastric H^+/Cl^- loss, volume contraction and hypokalaemia.
> - **First step: urine chloride** (expect < 25); **treat with 0.9 % saline + KCl**.

Q4. (Slide Case 4) A 70-year-old with sepsis, lactate 5: pH 7.20, $PaCO_2$ 25, HCO_3^- 10; Na 135, Cl 95. **A. HAGMA with appropriate respiratory compensation** · B. HAGMA without appropriate compensation · C. NAGMA with appropriate compensation · D. NAGMA without

> **Answer: A** (slide key, verified). AG = 30; Winter's = 23 ± 2 (21–25); actual 25 → appropriate.

Q5. A man drank windscreen-washer fluid. He has blurred vision, pH 7.10, AG 28 and a raised osmolar gap. Specific treatment? A. IV bicarbonate only · B. Isotonic saline · **C. Fomepizole + haemodialysis** · D. Acetazolamide · E. Paper-bag rebreathing

> **Answer: C** (slide 16). **Methanol** → formic acid (optic nerve toxicity). Fomepizole blocks alcohol dehydrogenase; haemodialysis removes the toxin.

Q6. A hypertensive woman has K^+ 2.9, HCO_3^- 33, pH 7.49 and **urine Cl^- 60 mEq/L**. She takes no diuretics. Likely cause, and will saline correct it? A. Vomiting; yes · **B. Primary hyperaldosteronism; no (chloride-resistant)** · C. Diarrhoea; yes · D. Addison's; no · E. Salicylates; yes

> **Answer: B** (slide 17). Chloride-resistant alkalosis (urine Cl^- > 40) with HTN + hypokalaemia → mineralocorticoid excess (link: Adrenal 1). Treat the cause.

Q7. ABG: pH 7.40, $PaCO_2$ 40, HCO_3^- 24; Na 140, Cl 100. Most accurate statement? A. Normal acid–base status · **B. HAGMA (AG 16) with a coexisting metabolic alkalosis** · C. Compensated respiratory acidosis · D. Lab error

> **Answer: B** (slide 22: "check the AG even in the absence of acidaemia").
> - AG = 16 (high) despite normal pH/HCO_3^-.
> - *[added]* Delta ratio = 4/0 → **metabolic alkalosis** offsets the acidosis (e.g. DKA with vomiting).

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Normal pH → no disorder	❌ **Calculate the AG**; mixed disorders can normalise pH
Compensation corrects pH fully	❌ **Never overcompensates**; overshoot = second disorder
$PaCO_2$ low in metabolic acidosis → fine	⚠️ Compare with **Winter's**; too low = **+ respiratory alkalosis**
Salicylates → pure metabolic acidosis	❌ **Mixed**: early respiratory alkalosis + late HAGMA
Acute vs chronic applies to metabolic disorders	❌ Only **respiratory** disorders have acute/chronic renal compensation; respiratory compensation of metabolic disorders is **immediate**
Case 3 "uncompensated"	⚠️ $PaCO_2$ 48 = **expected (47 ± 2)** → compensated; the slide key conflicts with its own formula
Diuretics only cause Cl-responsive alkalosis	❌ **Active use → urine Cl^- high** (resistant); remote use → low (responsive)
Every metabolic alkalosis → saline	❌ Only **Cl-responsive**; resistant (aldosterone, Cushing's) needs cause-directed treatment
Bicarbonate for all metabolic acidosis	❌ **Treat the cause**; HCO_3^- only if **severe**
High-flow O_2 in a hypercapnic COPD patient	❌ **Controlled O_2** (88–92 %)
AG uses K^+	⚠️ Standard AG = **Na − (Cl + HCO_3)** (6–12); with K^+ 10–16
Low albumin + normal AG = no gap acidosis	❌ *[added]* **Correct the AG for albumin**
Paper bag for every hyperventilating patient	❌ *[added]* Exclude **PE, MI, DKA, asthma** first
SMORE confusion	**Same** direction (pH & $PaCO_2$) = **Metabolic**; **Opposite** = **REspiratory**